

IEDA 4000H

Homework 1 Solutions

1. You are a CFA (chartered financial analyst). Betty has come to you because she needs help paying off her credit card bills. She owes the amounts on her credit cards shown in Table 1. Betty is willing to allocate up to \$6000 per month to pay off these credit cards. All cards must be paid off within 6 months. Betty's goal is to minimize the total of all her payments. To solve this problem, you must understand how interest on a loan works. To illustrate, suppose Betty pays \$6000 on BOCHK during month 1. Then her BOCHK balance at the beginning of month 2 is

$$10000 \times (1 + 0.005) - 6000 = 4050.$$

This follows because during month 1 Betty incurs $10000 \times 0.5\%$ in interest charges on her BOCHK card.

Table 1: Credit Card Bills

Bank	Balance(\$)	Monthly Rate(%)
BOCHK	10000	0.5
HSBC	15000	1.0
SC	5000	1.5

- (a) Formulate a linear program to minimize the total payments Betty makes over the next six months.
- (b) Solve the linear program from the previous part using the CVXPY library in Python.

Solution:

- (a) Let $i = 1, 2, 3$ index the credit cards, where $i = 1$ is BOCHK ($B_1 = 10000$, $r_1 = 0.005$), $i = 2$ is HSBC ($B_2 = 15000$, $r_2 = 0.01$), and $i = 3$ is SC ($B_3 = 5000$, $r_3 = 0.015$). Let $t = 1, 2, \dots, 6$ index the months. Let $x_{i,t} \geq 0$ be the payment allocated to card i in month t . The objective is to minimize the total payments:

$$\min \sum_{i=1}^3 \sum_{t=1}^6 x_{i,t},$$

subject to the monthly budget constraints:

$$\sum_{i=1}^3 x_{i,t} \leq 6000, \quad t = 1, 2, \dots, 6,$$

and the payoff constraints ensuring each card's balance is non-positive after 6 months:

$$\sum_{t=1}^6 x_{i,t}(1+r_i)^{6-t} \geq B_i(1+r_i)^6, \quad i = 1, 2, 3.$$

(b) Here is the Python code using the CVXPY library to solve the linear program:

```

1 import cvxpy as cp
2 import numpy as np
3
4 # Data
5 rates = np.array([0.005, 0.01, 0.015])
6 balances = np.array([10000, 15000, 5000])
7 n_cards = 3
8 n_months = 6
9 max_per_month = 6000
10
11 # Variables
12 x = cp.Variable((n_cards, n_months), nonneg=True)
13
14 # Objective
15 total_payments = cp.sum(x)
16 objective = cp.Minimize(total_payments)
17
18 # Constraints
19 constraints = []
20 # Monthly budget constraints
21 for t in range(n_months):
22     constraints.append(cp.sum(x[:, t]) <= max_per_month)
23
24 # Payoff constraints for each card
25 for i in range(n_cards):
26     r = rates[i]
27     B = balances[i]
28     grown = B * (1 + r)**n_months
29     weighted_sum = 0
30     for t in range(n_months):
31         power = n_months - (t + 1)
32         weighted_sum += x[i, t] * (1 + r)**power
33     constraints.append(weighted_sum >= grown)
34
35 # Problem

```

```

36 prob = cp.Problem(objective, constraints)
37
38 # Solve
39 prob.solve()
40
41 # Results
42 if prob.status == cp.OPTIMAL:
43     print("Status:", prob.status)
44     print(f"Minimal total payments: {prob.value:.2f}")
45     print("\nPayment schedule:")
46     for i in range(n_cards):
47         print(f"Card {i+1} (Balance: {balances[i]}, Rate: {rates[i]*100}%):")
48         for t in range(n_months):
49             print(f"    Month {t+1}: {x[i, t].value:.2f}")
50 else:
51     print("Status:", prob.status)
52     print("Solver failed to find an optimal solution.")

```

Running this code yields the following optimal solution:

- **Status:** optimal
- **Minimal total payments:** \$30713.87

Payment schedule:

- **Card 1 (Balance: \$10000, Rate: 0.5%):**
 - Month 1: \$0.00
 - Month 2: \$0.00
 - Month 3: \$0.00
 - Month 4: \$3524.57
 - Month 5: \$6000.00
 - Month 6: \$713.87
- **Card 2 (Balance: \$15000, Rate: 1.0%):**
 - Month 1: \$925.00
 - Month 2: \$6000.00
 - Month 3: \$6000.00
 - Month 4: \$2475.43

- Month 5: \$0.00
- Month 6: \$0.00
- **Card 3 (Balance: \$5000, Rate: 1.5%):**
 - Month 1: \$5075.00
 - Month 2: \$0.00
 - Month 3: \$0.00
 - Month 4: \$0.00
 - Month 5: \$0.00
 - Month 6: \$0.00

2. Let $S_{++}^n = \{X \in \mathbb{R}^{n \times n} \mid X^\top = X, X \succ 0\}$.

(a) Show that S_{++}^n is a convex set.

(b) Consider the function $f(X) = \log \det X$ with $\text{dom} f = S_{++}^n$. Show that f is concave.

Solution:

(a) For any $X, Y \in S_{++}^n$ and $\lambda \in [0, 1]$, we have

$$Z := \lambda X + (1 - \lambda)Y$$

is obviously symmetric and positive definite, i.e. $Z \in S_{++}^n$. Thus S_{++}^n is convex.

(b) For any $X \in S_{++}^n$, we define

$$g(t) := f(X + tV),$$

where $V \in \mathbb{R}^{n \times n}$ is symmetric and $t \in \{t \in \mathbb{R} \mid X + tV\}$. Since $X \in S_{++}^n$, then we have

$$\begin{aligned} X &= Q\Lambda Q^\top \\ &= Q\Lambda^{\frac{1}{2}}Q^\top Q\Lambda^{\frac{1}{2}}Q^\top \\ &:= X^{\frac{1}{2}}X^{\frac{1}{2}}. \end{aligned}$$

Therefore, we obtain

$$\begin{aligned} g(t) &= \log \det(X + tV) \\ &= \log \det \left(X^{\frac{1}{2}} \left(I + tX^{-\frac{1}{2}}VX^{-\frac{1}{2}} \right) X^{\frac{1}{2}} \right) \\ &= \log \det \left(X \left(I + tX^{-\frac{1}{2}}VX^{-\frac{1}{2}} \right) \right) \\ &= \log \det(X) + \log \det \left(I + tX^{-\frac{1}{2}}VX^{-\frac{1}{2}} \right) \\ &= \log \det(X) + \sum_{i=1}^n \log(1 + t\mu_i), \end{aligned}$$

where μ_i is the i -th eigenvalue of $X^{-\frac{1}{2}}VX^{-\frac{1}{2}}$. Hence, we have

$$g'(t) = \sum_{i=1}^n \frac{\mu_i}{1 + t\mu_i}, \quad g''(t) = -\sum_{i=1}^n \frac{\mu_i^2}{(1 + t\mu_i)^2} \leq 0.$$

This implies that $g(t)$ is concave. Therefore, f is concave.

3. Let Ω be a nonempty set, convex and bounded subset of $\mathbb{R}^n$. Define

$$\mu_{\Omega}(\mathbf{x}) := \sup\{\|\mathbf{x} - \boldsymbol{\omega}\| \mid \boldsymbol{\omega} \in \Omega\}, \quad \mathbf{x} \in \mathbb{R}^n.$$

Show that

- (a) μ_{Ω} is well defined, i.e. $\mu_{\Omega}(\mathbf{x}) < \infty$ for all $\mathbf{x} \in \mathbb{R}^n$.
- (b) μ_{Ω} is convex.

Solution:

- (a) Since Ω is bounded, there exists some $M > 0$ such that $\|\boldsymbol{\omega}\| \leq M$ for all $\boldsymbol{\omega} \in \Omega$.
For any $\mathbf{x} \in \mathbb{R}^n$, we have

$$\|\mathbf{x} - \boldsymbol{\omega}\| \leq \|\mathbf{x}\| + \|\boldsymbol{\omega}\| \leq \|\mathbf{x}\| + M.$$

Thus, $\mu_{\Omega}(\mathbf{x}) \leq \|\mathbf{x}\| + M < \infty$.

- (b) For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\lambda \in [0, 1]$, let $\mathbf{z} := \lambda\mathbf{x} + (1 - \lambda)\mathbf{y}$. For any $\epsilon > 0$, there exists $\boldsymbol{\omega}_z \in \Omega$ such that

$$\begin{aligned} \mu_{\Omega}(\mathbf{z}) - \epsilon &\leq \|\mathbf{z} - \boldsymbol{\omega}_z\| \\ &= \|\lambda(\mathbf{x} - \boldsymbol{\omega}_z) + (1 - \lambda)(\mathbf{y} - \boldsymbol{\omega}_z)\| \\ &\leq \lambda\|\mathbf{x} - \boldsymbol{\omega}_z\| + (1 - \lambda)\|\mathbf{y} - \boldsymbol{\omega}_z\| \\ &\leq \lambda\mu_{\Omega}(\mathbf{x}) + (1 - \lambda)\mu_{\Omega}(\mathbf{y}). \end{aligned}$$

The arbitrariness of ϵ implies that μ_{Ω} is convex.